MASTER'S RESEARCH PROPOSAL : SYSTEMS AND RETRIEVAL EFFICIENCY

Cascaded Adaptive Blended RAG: A Systems Extension for Efficient Multi-Field Hybrid Retrieval

BASE PAPER: Sawarkar et al. (IEEE MIPR 2024) **CORE FOCUS:** Pre-Retrieval Early Exit for Hybrid RAG

TARGET HARDWARE: Laptop and CPU Reproducible

01 BASE PAPER FOUNDATION AND THE LATENCY BOTTLENECK

What Blended RAG showed: Sawarkar et al. (IBM Research, 2024) proved that blending dense vector embeddings, learned sparse representations (SPLADE), and BM25 across document fields (title, abstract, text) significantly beats single-retriever systems. On standard benchmarks, it achieved **68% F1 on SQuAD**, **42% Exact Match on NQ**, and **88.8% retrieval accuracy on NQ** without task-specific fine-tuning.

THE STATIC HYBRID PENALTY — RESEARCH PROBLEM

Blended RAG runs all three retrieval models (Dense vector index, Sparse SPLADE model, and BM25 lexical index) for 100% of incoming queries. For simple factual queries where keywords already identify the right passage, running expensive neural vector search is redundant compute. Easy queries pay the exact same 30 to 80 ms latency penalty as complex multi-hop queries.

02 PROPOSED SYSTEMS EXTENSION: CASCADED BLENDED RAG

DEFENSIBLE SYSTEMS POSITIONING

We frame this work directly as **the first adaptive systems extension of Blended RAG**. We do not claim to invent adaptive RAG as a general concept. Instead, we remove Blended RAG's static execution bottleneck using a fast sequential cascade while preserving its multi-field metadata design.

CONCRETE EFFICIENCY TARGETS

35–50% Faster · ≥ 96% Acc.

Bypasses neural vector search on **40% to 60% of queries** via BM25 early exit, while maintaining ≥ 96% of static Blended RAG Recall@10, MRR, and downstream QA generation scores.

03 ARCHITECTURAL SPECIFICATION: THE FOUR TECHNICAL PILLARS

TIER 0: FEATURE CHECK — < 1 ms (CPU)

Pre-Retrieval Specificity Check

Extracts lexical specificity directly from query text before index lookup: query length, maximum inverse document frequency (MaxIDF), and rare token ratio.

TIER 1: BM25 FAST-PASS — 2 to 5 ms (CPU)

Confidence Margin Early Exit

Runs fast lexical search. If top score $s_1 \geq \tau_high$ or score margin $\Delta = s_1 - s_2 \geq \tau_gap$, halts and returns candidates immediately. **Skips 100% of neural vector compute.**

TIER 2: BLENDED FALLBACK — 30 to 80 ms

3-Way Hybrid Fusion (RRF)

When lexical confidence is low ($\Delta < \tau_gap$), escalates to full Blended RAG: Dense MiniLM-L6 + SPLADE sparse term expansion + BM25, merged via Reciprocal Rank Fusion.

CROSS-TIER: METADATA — Zero Extra Overhead

Best Fields Matching

Applies Blended RAG's core method: searching across structured fields (title, abstract, text) so scientific corpora like SciFact retain high precision across both tiers.

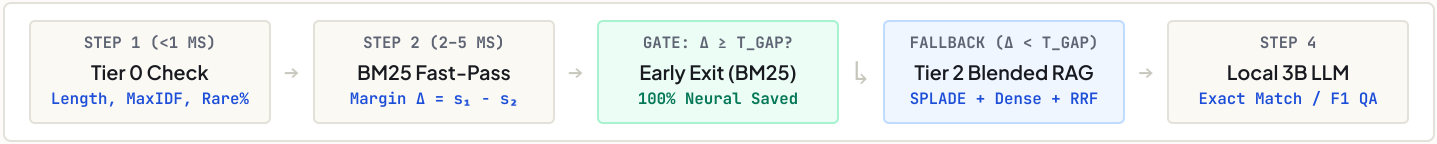
SEQUENTIAL DECISION RULE

```
Route(Q) = BM25_Exit IF [s₁ ≥ τ_high] OR [Δ ≥ τ_gap AND MaxIDF(Q) ≥ τ_idf], ELSE Tier2_BlendedRAG
```

04 LITERATURE POSITIONING AND PRIOR-ART DIFFERENTIATION (2024–2026)

Recent literature adapts different parts of RAG (query complexity, fusion weights, or passage depth). Our system is distinguished by **exiting early before heavy neural search** while preserving multi-field hybrid fusion:

APPROACH / PAPER	WHAT ADAPTS	DECISION POINT	SAVES NEURAL COMPUTE?	MULTI-FIELD RRF?	KEY DIFFERENTIATION
Blended RAG *(Sawarkar 2024)*	None (Static)	Static (100% queries)	No (0% saved)	Yes	Target base paper; heavy static execution penalty
Adaptive-RAG *(Jeong 2024)*	Retrieval strategy	Query complexity	Partial	No	Routes single vs multi-step; not hybrid early exit
DAT *(Hsu & Tzeng 2025)*	Sparse-dense weight (α)	Per-query predictor	No (Runs all first)	No	Reweights scores; saves zero retrieval latency
Query-Adaptive *(Posokhov 2026)*	Sparse-dense weight	Latency predictor	No (Runs both)	No	Continuous α-tuning; cannot early-exit lexical path
SAGE *(Raza et al. 2026)*	Passage budget k	Lightweight probe	Depth only	No	Adapts retrieved depth, not retriever modalities
Our Cascaded Blended RAG	**Sequential Early Exit**	**Pre-Retrieval + BM25**	Yes (40–60% skipped)	Yes	**First adaptive systems extension of Blended RAG**



EXPERIMENTAL PROTOCOL AND BASELINES

- **Datasets:** BEIR SciFact (5,183 scientific papers, 300 test queries) and Natural Questions (NQ) subset for direct comparison with Sawarkar et al.
- **Baselines:** (1) Pure BM25, (2) Dense MiniLM-L6, (3) Sparse SPLADE, (4) Static Blended RAG, (5) Adaptive Baselines (Adaptive-RAG and DAT).
- **System Metrics:** Mean and tail latency (P50, P90, P99 ms), and percentage of neural vector searches avoided.
- **Quality Metrics:** Recall@5, Recall@10, MRR@10, nDCG@10, plus Exact Match and F1 using a local 3B generator.

RISK MANAGEMENT AND BACKUP PLAN

- **Handling router errors (Wu & You, 2026):** Adaptive routers can misclassify ambiguous queries. We calibrate conservative thresholds (`τ_gap, τ_high`) via 5-fold cross-validation.
- **Backup plan 1:** If early exit drops recall by over 2%, pivot the router to dynamic alpha weighting for 3-way Blended RRF.
- **Backup plan 2:** Shift to adaptive top-k depth truncation for Blended RAG (following SAGE and CAR).
- **Defense value:** Mapping the exact boundary where lexical confidence works provides direct systems value for production RAG.

KEY ACADEMIC REFERENCES (2024–2026)

Sawarkar, Mangal, & Solanki (2024). Blended RAG: Improving RAG with Semantic Search and Hybrid Retrievers. *IEEE MIPR.*

Hsu & Tzeng (2025). DAT: Dynamic Alpha Tuning for Hybrid Retrieval in RAG. *arXiv:2503.23013.*

Wu & You (2026). Evidence-Unit Fairness and Limits of Query-Adaptive Fusion. *arXiv:2608.00183.*

Jeong et al. (2024). Adaptive-RAG: Learning to Adapt Retrieval-Augmented LLMs through Question Complexity. *NAACL 2024.*

Raza, Yang, & Srinivasan (2026). SAGE: SLO-Aware Adaptive Retrieval for Production RAG Systems. *IEEE CoDIT 2026.*

Posokhov et al. (2026). Query-Adaptive Hybrid Search. *Mach. Learn. Knowl. Extr.*, MDPI.